

Scanning the human proteome for calmodulin-binding proteins

The calcium ion (Ca^{2+}) is a ubiquitous second messenger that is crucial for the regulation of a wide variety of cellular processes. The diverse transient signals transduced by Ca^{2+} are mediated by intracellular Ca^{2+} -binding proteins, also known as Ca^{2+} sensors. A key obstacle to studying many Ca^{2+} -sensing proteins is the difficulty in identifying the numerous downstream target interactions that respond to Ca^{2+} -induced conformational changes. Among a number of Ca^{2+} sensors in the eukaryotic cell, calmodulin (CaM) is the most widespread and the best studied. Employing the mRNA display technique, we have scanned the human proteome for CaM-binding proteins and have identified and characterized a large number of both known and previously uncharacterized proteins that interact with CaM in a Ca^{2+} -dependent manner. The interactions of several identified proteins with Ca^{2+} /CaM were confirmed by using pull-down assays and coimmunoprecipitation. Many of the CaM-binding proteins identified belong to protein families such as the DEAD/H box proteins, ribosomal proteins, proteasome 26S subunits, and deubiquitinating enzymes, suggesting the possible involvement of Ca^{2+} /CaM in different signaling pathways. The selection method described herein could be used to identify the binding partners of other calcium sensors on the proteome-wide scale.